



Programme: B E – Information Science & Engineering

Internal Assessment – I

TERM :15/04/2024 to 22/07/2024	COURSE NAME: Database Management Systems
DATE :31/05/2024 TIME: 2.00 PM -3.00 PM	COURSE CODE : IS44
SEMESTER: IV	MAX MARKS: 30
	PORTIONS : L 1 – L1 6



Mobile Phones are banned

Instructions to Candidates: Answer any TWO full questions.

Marks: 15x2=30

Q. NO.	Questions	Blooms Levels (L1 to L6)*	CO	Marks																								
1.a	A retail chain, "ShopSmart," operates over 100 stores across the country. Each store needs to maintain its own data. Explain how the database approach is advantageous over the traditional file system for this scenario.	L4	CO1	7																								
b	List and explain any two characteristics of relational model with an example.	L2	CO2	4																								
c	Consider the below table, write the SQL commands to perform the below operations. Table 1 - EMPLOYEE <table><tr><th>EmployeeID</th><th>FirstName</th><th>LastName</th><th>Age</th><th>Department</th><th>Salary</th></tr><tr><td>1</td><td>John</td><td>Doe</td><td>30</td><td>HR</td><td>60000</td></tr><tr><td>2</td><td>Jane</td><td>Smith</td><td>35</td><td>Finance</td><td>70000</td></tr><tr><td>3</td><td>Emily</td><td>Johnson</td><td>40</td><td>IT</td><td>80000</td></tr></table> i) Create a table with necessary constraints. ii) Change the department of <i>Jane Smith</i> to 'IT' iii) Delete the <i>Age</i> column from the Employees table. iv) Add a new column called <i>Email</i>	EmployeeID	FirstName	LastName	Age	Department	Salary	1	John	Doe	30	HR	60000	2	Jane	Smith	35	Finance	70000	3	Emily	Johnson	40	IT	80000	L3	CO3	4
EmployeeID	FirstName	LastName	Age	Department	Salary																							
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2.a	A database is being constructed to keep track of the teams and games of a sports league. A team has a number of players, not all of whom participate in each game. It is desired to keep track of players participating in each game for each team, the positions they played in that game and the result of the game. Design an ER diagram for this application, state if any assumptions you make.	L3	CO1	7																								
b	What is Foreign Key? Explain its importance in databases with an example.	L2	CO2	4																								
c	With reference to a Table 1 – EMPLOYEE (Question 1c.), Write the SQL statements to i) Calculate the total salary expenditure for all employees ii) Select employees sorted by salary in descending order iii) Select the top 3 highest paid employees	L3	CO3	4																								
3.a	How do you achieve logical and physical data independence using three schema architecture?	L2	CO1	7																								
b	List the relational database Constraints. With an example explain when Entity Integrity Constraint is violated.	L2	CO2	4																								
c	Explain different ways of using INSERT command in SQL with one example for each.	L2	CO3	4																								

* L1 – Remember, L2 – Understand, L3- Apply, L4- Analyze, L5-Evaluate, L6-Create